

①

$$L = \{a^i b^j c^k \mid i=j \text{ o } j \leq k\}$$

$$S \rightarrow AE \mid B$$

$$A \rightarrow aAb \mid \lambda \quad \leftarrow N(a) \neq N(b)$$

$$B \rightarrow CD$$

$$C \rightarrow aC \mid \lambda \quad \alpha = aA$$

$$D \rightarrow bDc \mid E$$

$$E \rightarrow cE \mid \lambda$$

$$AE = N(a) \neq N(b) \quad \uparrow \text{Minúscula}$$

$$B = N(c) \geq N(b)$$

Paso 1: Eliminar producciones unitarias

- Localizar símbolos anulables

$$A_0 = \{E, C, A\}$$

$$A_1 = \{E, C, D, A\}$$

$$A_2 = \{A, B, C, D, E\}$$

$$A_3 = \{S, A, B, C, D, E\}$$

Reescribimos las normas

$$S \rightarrow A \mid E \mid AE \mid B \mid \lambda \quad \leftarrow \lambda \in L$$

$$A \rightarrow ab \mid aAb$$

$$B \rightarrow C \mid D \mid CD$$

$$C \rightarrow a \mid aC \mid \lambda$$

$$D \rightarrow bc \mid bDc \mid E$$

$$E \rightarrow c \mid \lambda \mid cE \quad \uparrow \text{absurdo}$$

Paso 3: Eliminar símbolos inútiles

1º Generadores:

$$G_0 = \{a, b, c\}$$

$$G_1 = \{A, C, D, E, a, b\}$$

$$G_2 = \{A, B, C, D, E, a, b, c\}$$

2º Alcanzables

$$A_0 = \{S\}$$

$$A_1 = \{S, A, C, D, E\}$$

$$A_2 = \{S, A, C, D, E\}$$

Símbolo inútil = B

Reescribo normas:

$$S \rightarrow ab \mid aAb \mid a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD \mid AE$$

$$A \rightarrow ab \mid aAb$$

$$C \rightarrow a \mid aC$$

$$D \rightarrow bc \mid bDc \mid c \mid cE$$

$$E \rightarrow c \mid cE$$

Paso 2: Identificar pares unitarios:

$$\checkmark (A, A)$$

$$\checkmark (B, B) \quad (B, C) \quad (B, D) \quad (B, E)$$

$$\checkmark (C, C)$$

$$\checkmark (D, D) \quad (D, E)$$

$$\checkmark (E, E)$$

$$(S, S) \quad (S, A) \quad (S, E) \quad (S, B) \quad (S, D)$$

Reescribimos normas:

$$S \rightarrow ab \mid aAb \mid a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD \mid AE$$

$$A \rightarrow ab \mid aAb$$

$$B \rightarrow a \mid aC \mid bc \mid bDc \mid c \mid cE \mid CD$$

$$C \rightarrow a \mid aC$$

$$D \rightarrow bc \mid bDc \mid c \mid cE$$

$$E \rightarrow c \mid cE$$

②

$$L = \{a^i b^j a^k \mid j = i+k; i, j, k \geq 0\}$$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid \lambda$$

$$B \rightarrow bBa \mid \lambda$$

Paso 2: Eliminar proyecciones con λ

Buscar anulables

$$A_0 = \{A, B\}$$

$$A_1 = \{A, B, S\} \rightarrow \text{Todos anulables}$$

Nueva gramática

$$S \rightarrow AB \mid \lambda \mid A \mid B$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow bBa \mid ba$$

Paso 1: Eliminar símbolos inútiles

Alcanzables

$$A_0 = \{S\} \quad A_1 = \{S, AB\} \quad A_2 = \{S, A, B\}$$

Generadores

$$G_0 = \{A, B\} \quad G_1 = \{A, B, S\}$$

Son todos generadores y alcanzables, por lo tanto no hay símbolos inútiles

Paso 3: Eliminar proyecciones unitarias

Por unitarios

$$(A, A)$$

$$(B, B)$$

$$(S, S) \quad (S, A) \quad (S, B)$$

Nueva gramática

$$S \rightarrow AB \mid \lambda \mid aAb \mid ab \mid bBa \mid ba$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow bBa \mid ba$$

③

$$S \rightarrow XaX \mid YbY$$

$$X \rightarrow XaXbX \mid XbXaX \mid aX \mid \lambda$$

$$Y \rightarrow YaYbY \mid YbYaY \mid bY \mid \lambda$$

Paso 2: Eliminar proyecciones con λ

$$A_0 = \{X, Y\} \quad A_1 = \{X, Y\}$$

Paso 1: No hay símbolos inútiles

$\forall \text{ símbolo } \in G$ es generador y alcanzable

Paso 3: Eliminar producciones unitarias, pues estamos de suerte porque no hay

$$X \rightarrow ab \mid Xab \mid aXb \mid abX \mid XaXb \mid aXbX \mid XabX \mid XaXbX \mid a \mid aX$$

$$ba \mid Xba \mid bXa \mid baX \mid XbXa \mid bXaX \mid XbaX \mid XbXaX$$

$$Y \rightarrow \text{Igual cambiando la } X \text{ por la } Y \text{ y } b \mid bY$$

$$S \rightarrow a \mid Xa \mid aX \mid XaX \mid b \mid Yb \mid bY \mid YbY$$

$$④ L = \{a^i b^j c^k \mid k \neq i+j\}$$

$k < i+j \quad k > i+j$

$$S \rightarrow P \mid Q$$

$$P \rightarrow A \mid B \mid AB \mid W$$

$$W \rightarrow AWc \mid X \mid a$$

$$X \rightarrow BXc \mid b$$

$$Q \rightarrow C \mid R$$

$$R \rightarrow aRC \mid Y \mid c$$

$$Y \rightarrow bYC \mid c$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$C \rightarrow Cc \mid c$$

Paso 2: Eliminar proyecciones con λ

FÁCIL, NO HAY

$$S \rightarrow Aa \mid Bb \mid Cc \mid AWc \mid BXc \mid aRC \mid bYC \mid a \mid b \mid c$$

$$P \rightarrow Aa \mid Bb \mid AWc \mid BXc \mid b \mid a$$

$$W \rightarrow AWc \mid BXc \mid b \mid a$$

$$X \rightarrow BXc \mid b$$

$$Q \rightarrow Cc \mid aRC \mid bYC \mid c$$

$$R \rightarrow aRC \mid bYC \mid c$$

$$Y \rightarrow bYC \mid c$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$C \rightarrow Cc \mid c$$

Paso 1: Eliminar símbolos inútiles

Todos son generadores

$$\begin{cases} G_0 = \{W, X, R, Y, A, B, C\} \\ G_1 = \{P, Q, W, X, R, Y, A, B, C\} \\ G_2 = S + G_1 \\ G_3 = G_2 \end{cases}$$

Todos son alcanzables

$$\begin{cases} A_0 = \{S\} \\ A_1 = A_0 + \{P, Q\} \\ A_2 = A_1 + \{W, R, A, B, C\} \\ A_3 = A_2 + \{X, Y\} \\ A_4 = A_3 \end{cases}$$

Paso 3: Eliminar proyecciones unitarias

(S, S)	$(S, P)(S, Q)$	$(S, C)(S, R)$	$(S, A)(S, B)(S, W)$	(S, W)	(S, Y)
(P, P)	$(P, A)(P, B)(P, W)$	(P, W)			
(W, W)	(W, X)				
(X, X)					
(Q, Q)	$(Q, C)(Q, R)$	(Q, Y)			
(R, R)	(R, Y)				
(Y, Y)					
	L_1	L_2	L_3		

⑤ $L = \{ a^i b^j c^k d^m \mid i \geq m \text{ y } j > k \}$

Paso 1: Símbolos inútiles
No hay símbolos inútiles

$$S \rightarrow aSD \mid AP$$

$$P \rightarrow bPC \mid B$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$D \rightarrow d \mid \lambda$$

$$C \rightarrow c \mid \lambda$$

Paso 2: Eliminar proyecciones con λ

$$S \rightarrow aS \mid aSd \mid AP$$

$$P \rightarrow bP \mid bPc \mid B$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

Paso 3: Eliminar proyecciones unitarias
(P, B)

$$\begin{aligned} S &\rightarrow aSd \mid AP \\ P &\rightarrow bP \mid bPc \mid Bb \mid b \\ A &\rightarrow Aa \mid a \\ B &\rightarrow Bb \mid b \end{aligned}$$

⑥ $L = \{ a^i (b+c)^k a^j \mid k \neq i+j, i, j, k \geq 0 \}$

$$S \rightarrow APQ \mid PQA \mid APQA \mid PBO$$

$$P \rightarrow aPZ \mid \lambda$$

$$Q \rightarrow ZYa \mid \lambda$$

$$Z \rightarrow b \mid c$$

$$B \rightarrow bB \mid b \mid cB \mid c$$

$$A \rightarrow Aa \mid a$$

Paso 1: Eliminar símbolos inútiles

No hay $\forall S \in G$ S es Generador y Alcanzable

Paso 2: Eliminar proyecciones con λ

$$S \rightarrow A \mid AP \mid APQ \mid AQ \mid PA \mid QA \mid PQA \mid APQA \mid PB \mid BQ \mid PBQ \mid B \mid APA \mid AQA$$

$$P \rightarrow aZ \mid aPZ$$

$$Q \rightarrow ZYa \mid Za$$

$$Z \rightarrow b \mid c$$

$$B \rightarrow bB \mid b \mid cB \mid c$$

Paso 3: Eliminar proyecciones unitarias

$$\begin{aligned} S &\rightarrow Aa \mid a \mid AP \mid APQ \mid AQ \mid PA \mid QA \mid PQA \mid APQA \mid PB \mid BQ \mid PBQ \mid bB \mid b \mid cB \mid c \mid APA \mid AQA \\ P &\rightarrow aZ \mid aPZ \\ Q &\rightarrow ZYa \mid Za \\ Z &\rightarrow b \mid c \\ B &\rightarrow bB \mid b \mid cB \mid c \end{aligned}$$

⑦ $L = \{ a^i b^j c^k a^m b^p \mid i \geq j+k \text{ y } m=p \}$

$$S \rightarrow PQ$$

$$P \rightarrow aPc \mid R$$

$$R \rightarrow aRb \mid \lambda$$

$$Q \rightarrow aQb \mid \lambda$$

Paso 1: Símbolos inútiles

$\forall S \in G$

$S \in \{\text{Generadores}\}$

y

$S \in \{\text{Alcanzables}\}$

Paso 3: Elimine proyecciones unitarias

Paso 2: Eliminar proy con λ

$$S \rightarrow P \mid Q \mid PQ \mid \lambda$$

$$P \rightarrow ac \mid aPc \mid R$$

$$R \rightarrow ab \mid aRb$$

$$Q \rightarrow ab \mid aQb$$

$$S \rightarrow ac \mid aPc \mid ab \mid aRb \mid aQb \mid \lambda \mid PQ$$

$$P \rightarrow ac \mid aPc \mid ab \mid aRb$$

$$R \rightarrow ab \mid aRb$$

$$Q \rightarrow ab \mid aQb$$

$$\textcircled{8} \quad L = \{ a^i b^j c^k d^m \mid i+k \leq j+m \text{ o } i+j+k \leq m \quad i, j, k, m \geq 0 \}$$

$$S \rightarrow E/F$$

$$\left. \begin{array}{l} E \rightarrow aEd \mid \lambda \\ J \rightarrow bJd \mid K \\ K \rightarrow cKd \mid \lambda \end{array} \right\} i+j+k \leq m$$

$$\left. \begin{array}{l} F \rightarrow aFd \mid LPM \\ L \rightarrow aLb \mid \lambda \\ M \rightarrow cMd \mid \lambda \\ P \rightarrow bPc \mid \lambda \end{array} \right\} i+k \leq j+m$$

Paso 1: Eliminar símbolos inútiles

$$\forall S \in G$$

S es generador

S es alcanzable

Paso 2: Eliminar proyecciones con λ

$$F \rightarrow ad \mid aFd \mid L \mid P \mid M \mid LP \mid LM \mid PM \mid LPM$$

$$L \rightarrow aLb \mid ab$$

$$M \rightarrow aMc \mid ac$$

$$P \rightarrow bPc \mid bc$$

Paso 3: Eliminar proyecciones unitarias

$$E \rightarrow ad \mid aEd \mid bJd \mid bd \mid cKd \mid cd$$

$$J \rightarrow bJd \mid bd \mid cKd \mid cd$$

$$K \rightarrow cKd \mid cd$$

$$F \rightarrow ad \mid aFd \mid aLb \mid ab \mid aMc \mid ac \mid bPc \mid bc \mid LP \mid LM \mid PM \mid LPM$$

$$L \rightarrow aLb \mid ab$$

$$M \rightarrow aMc \mid ac$$

$$P \rightarrow bPc \mid bc$$

$$S \rightarrow ad \mid bcl \mid ac \mid ab \mid aEd \mid bJd \mid cKd \mid aFd \mid aLb \mid aMc$$

$$bPc \mid LP \mid LM \mid PM \mid LPM$$